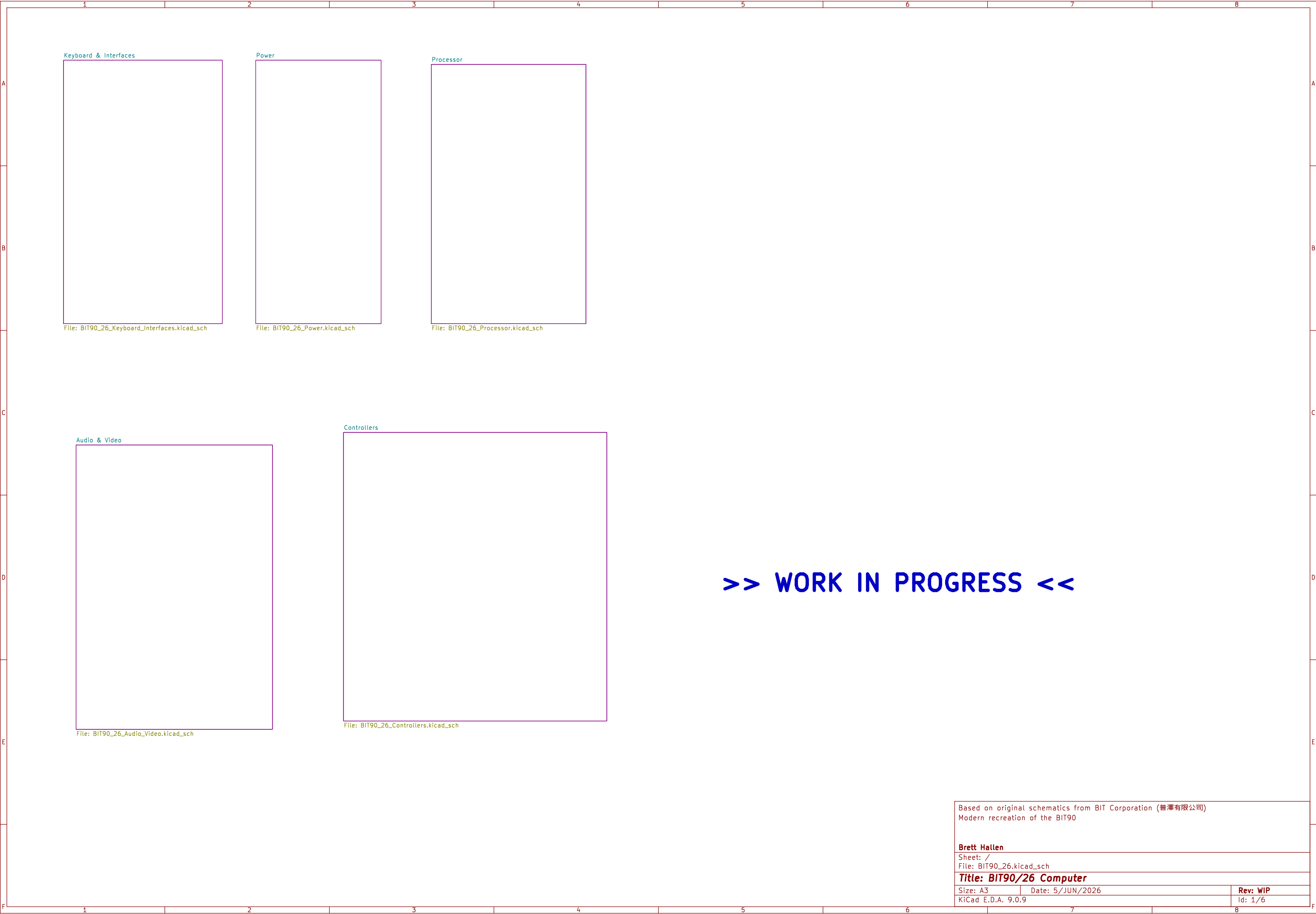
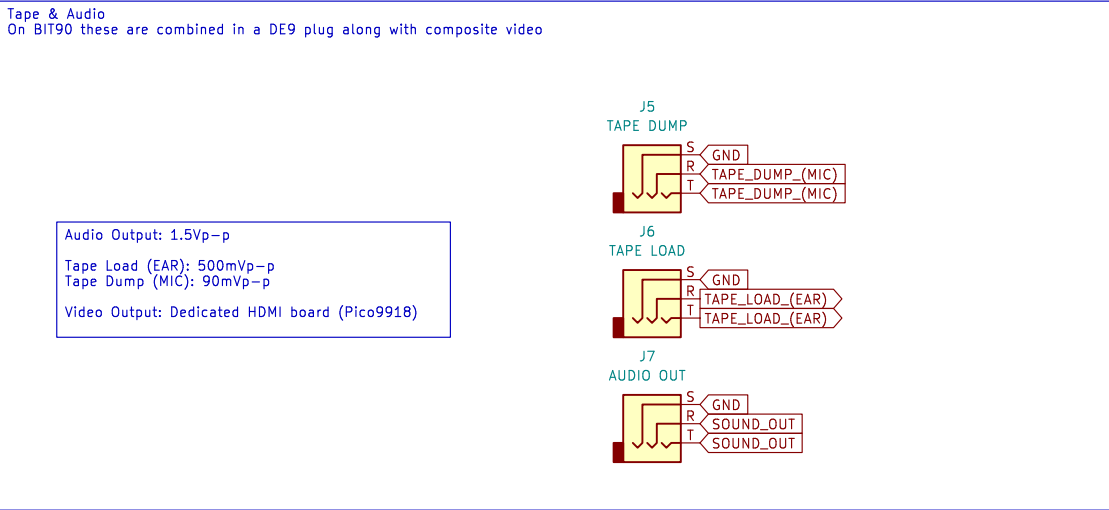
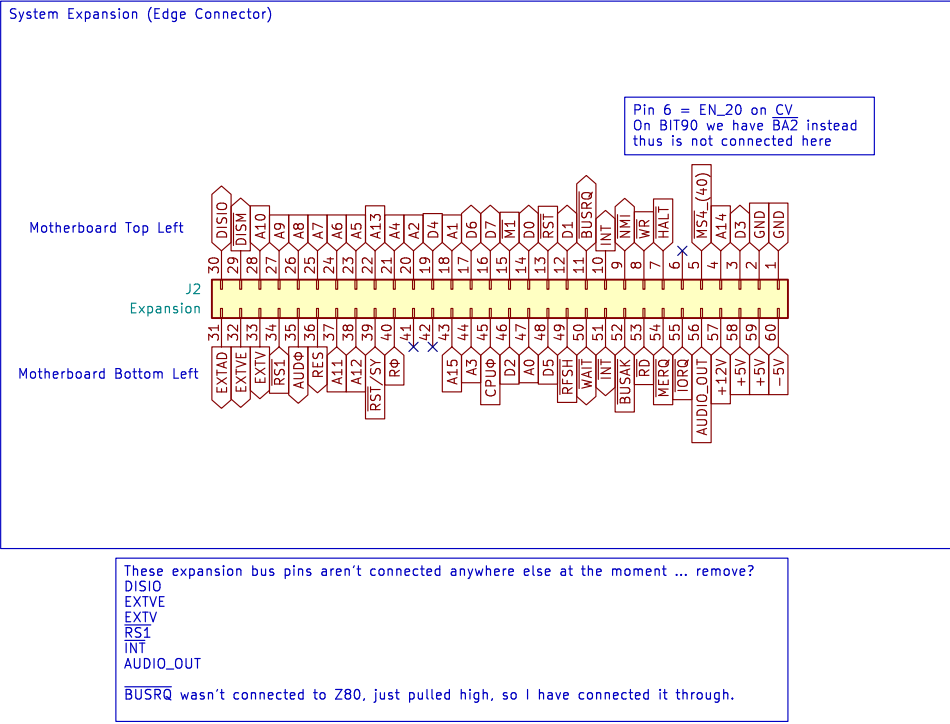
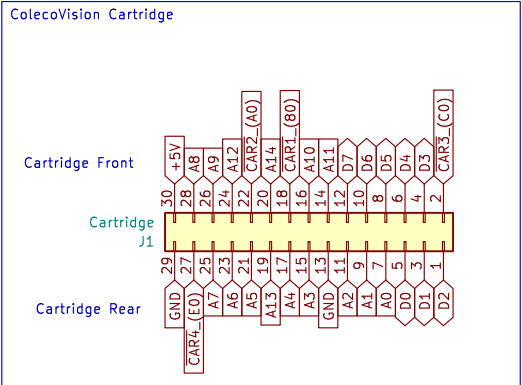
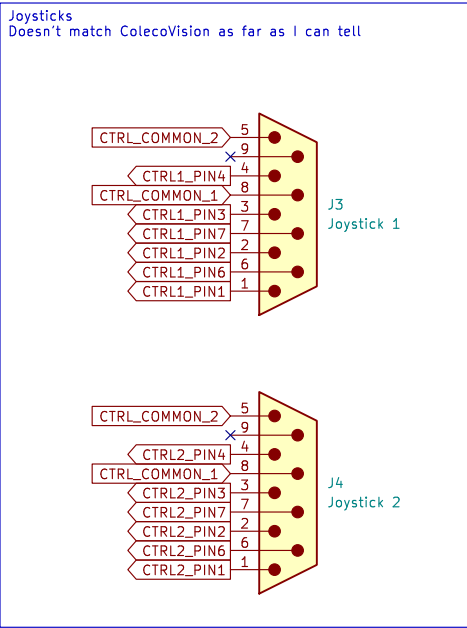
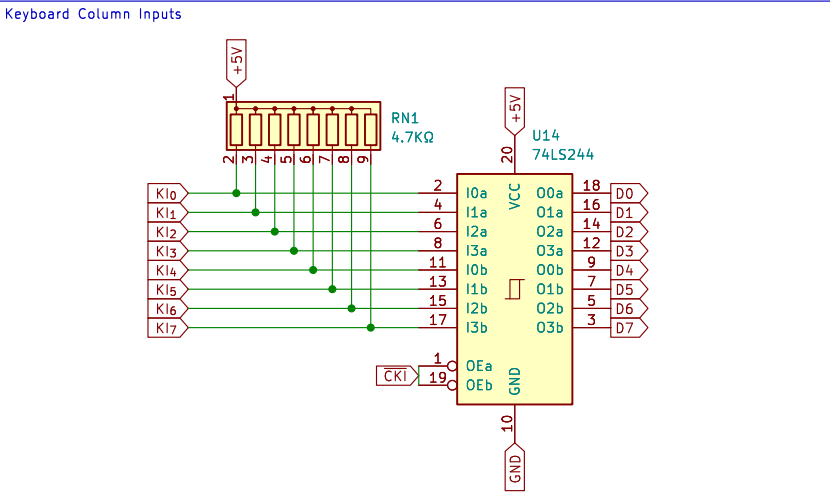
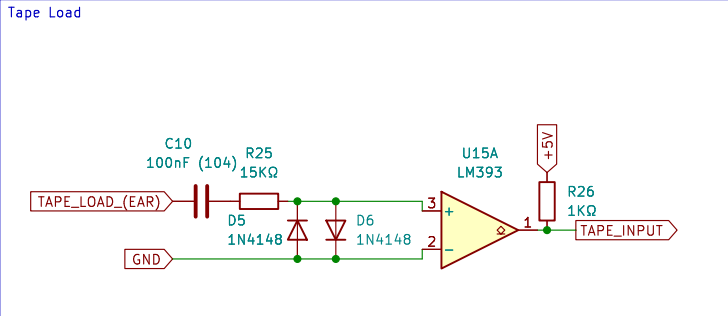
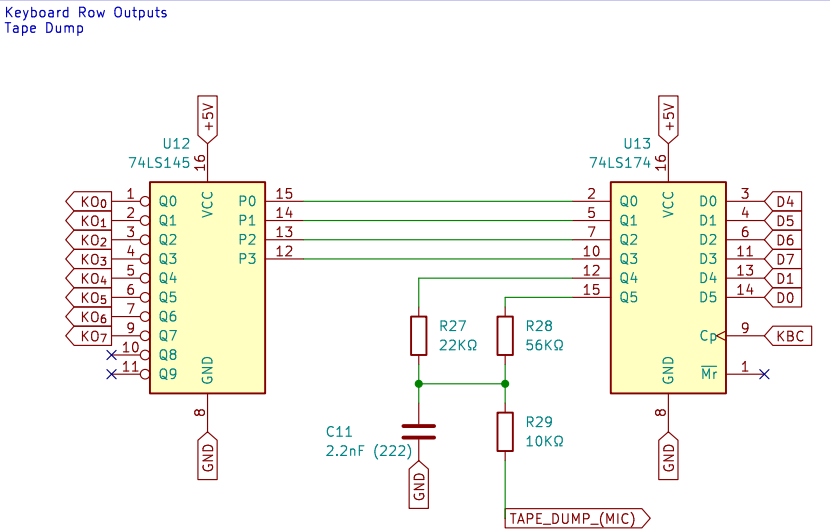
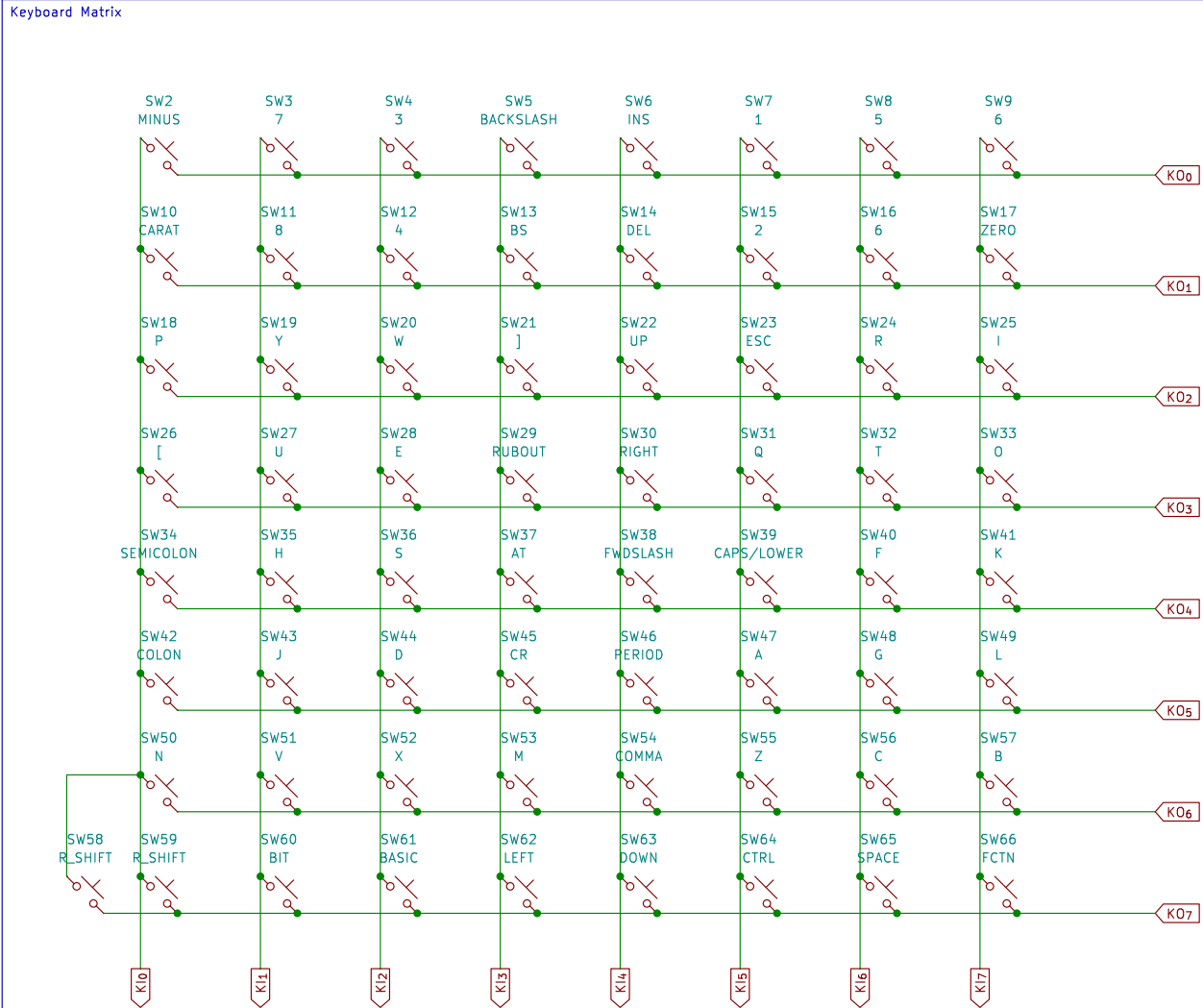
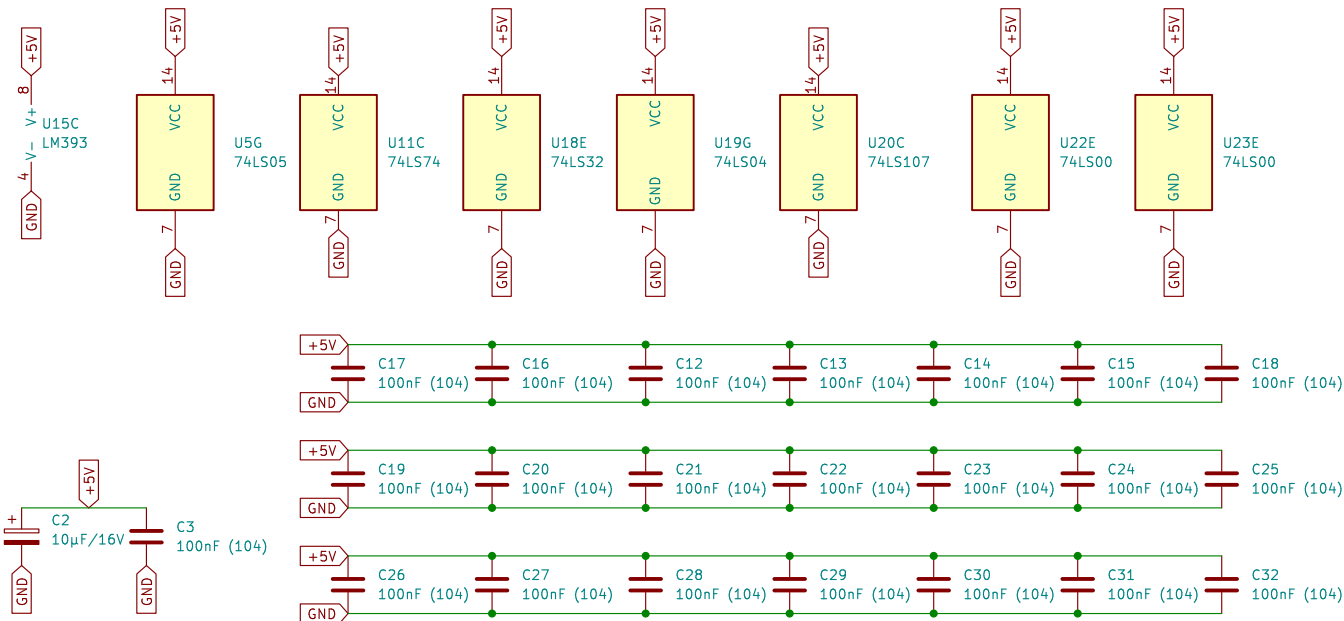


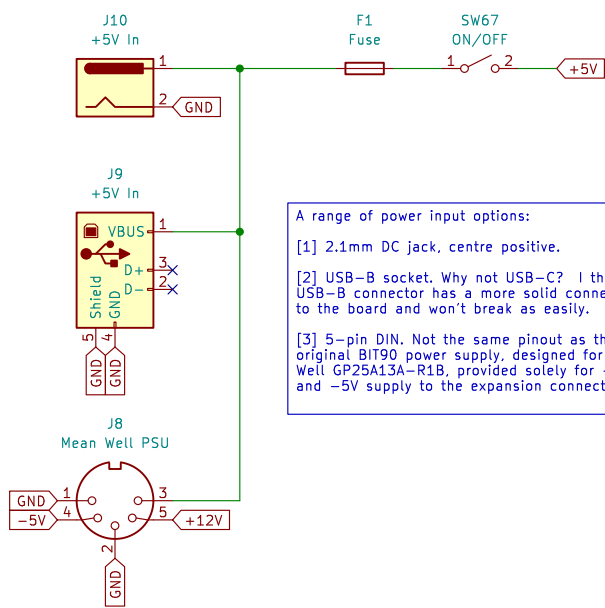




Power & Decoupling



Power Input
+5VDC



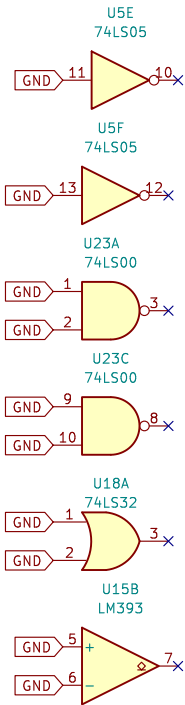
A range of power input options:

[1] 2.1mm DC jack, centre positive.

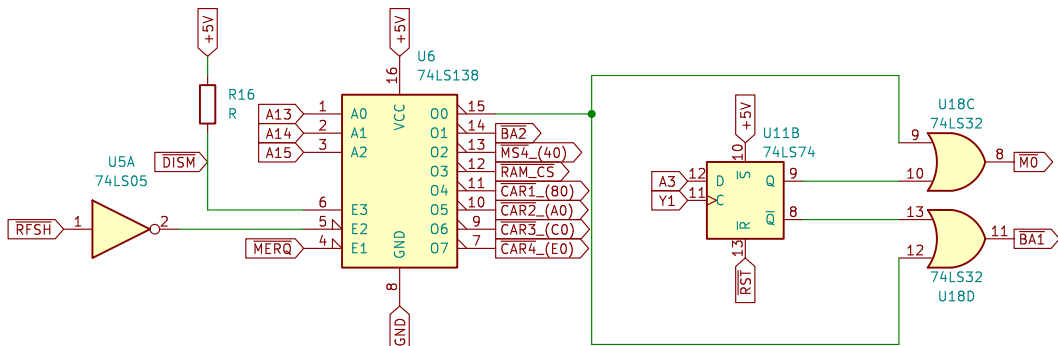
[2] USB-B socket. Why not USB-C? I think the USB-B connector has a more solid connection to the board and won't break as easily.

[3] 5-pin DIN. Not the same pinout as the original BIT90 power supply, designed for a Mean Well GP25A13A-R1B, provided solely for +12V and -5V supply to the expansion connector.

Spare Gates

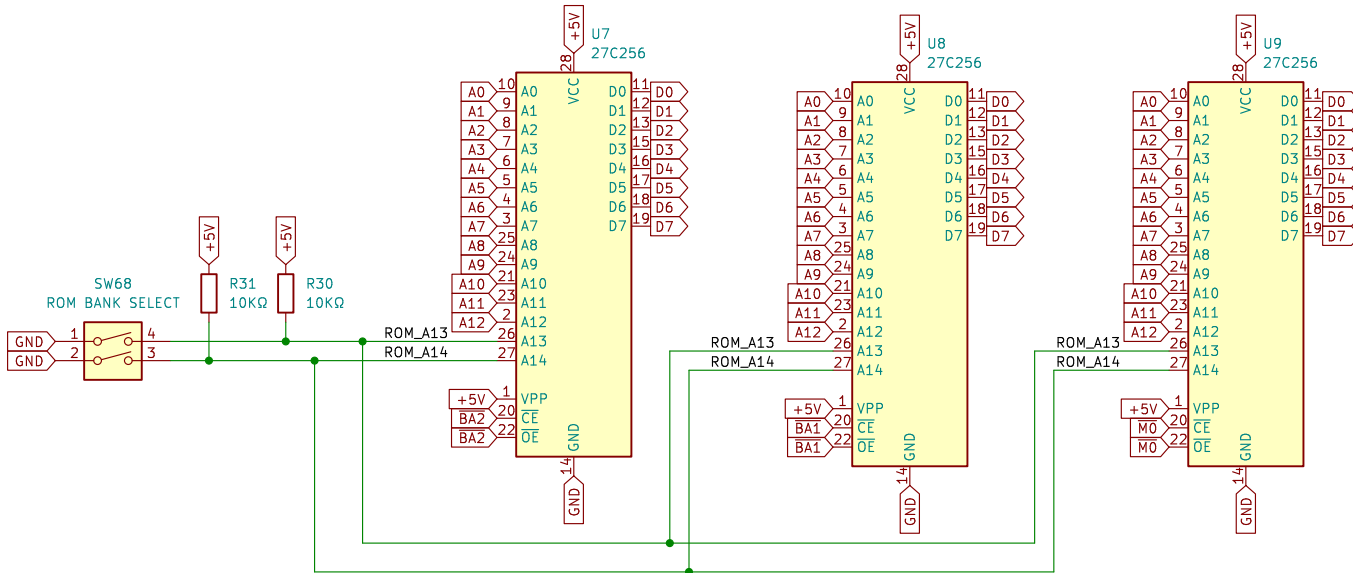


Chip Select

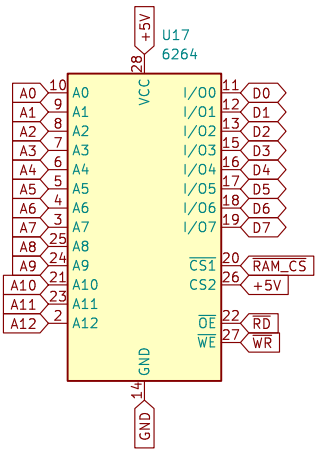


MO	ColecoVision BIOS
BA1 & BA2	BASIC
RAM_CS	System RAM
CAR1 to CAR4	Cartridge
MS4	Expansion

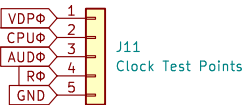
System ROMs
BASIC1, BASIC2, ColecoVision BIOS
Original ROMs are 8KB
Using 27C256 allows four banks that can be switched



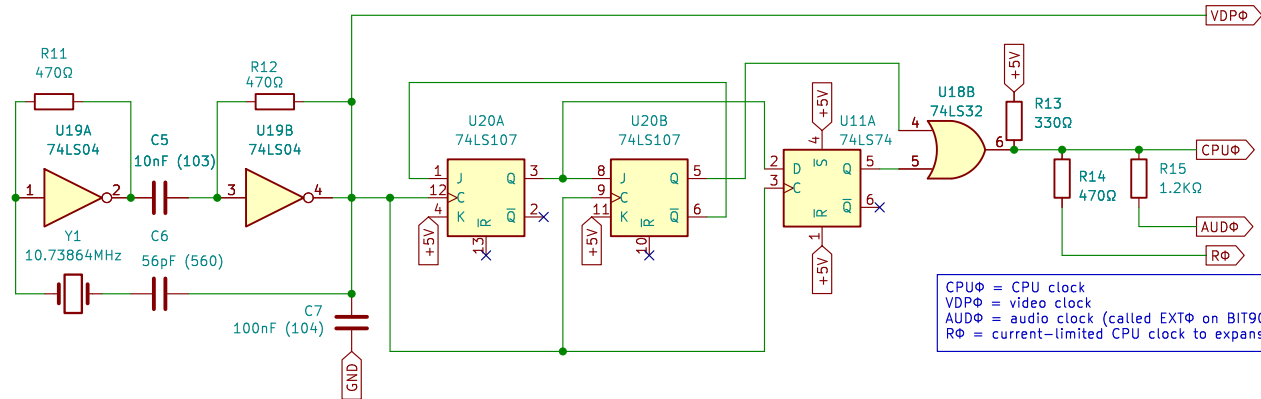
System RAM
BIT90 had 2KB originally in an 8KB block
Let's upgrade to 8KB
Upgrading further to 32KB will require additional logic
Let's see if this works first ...



Clock Test Points

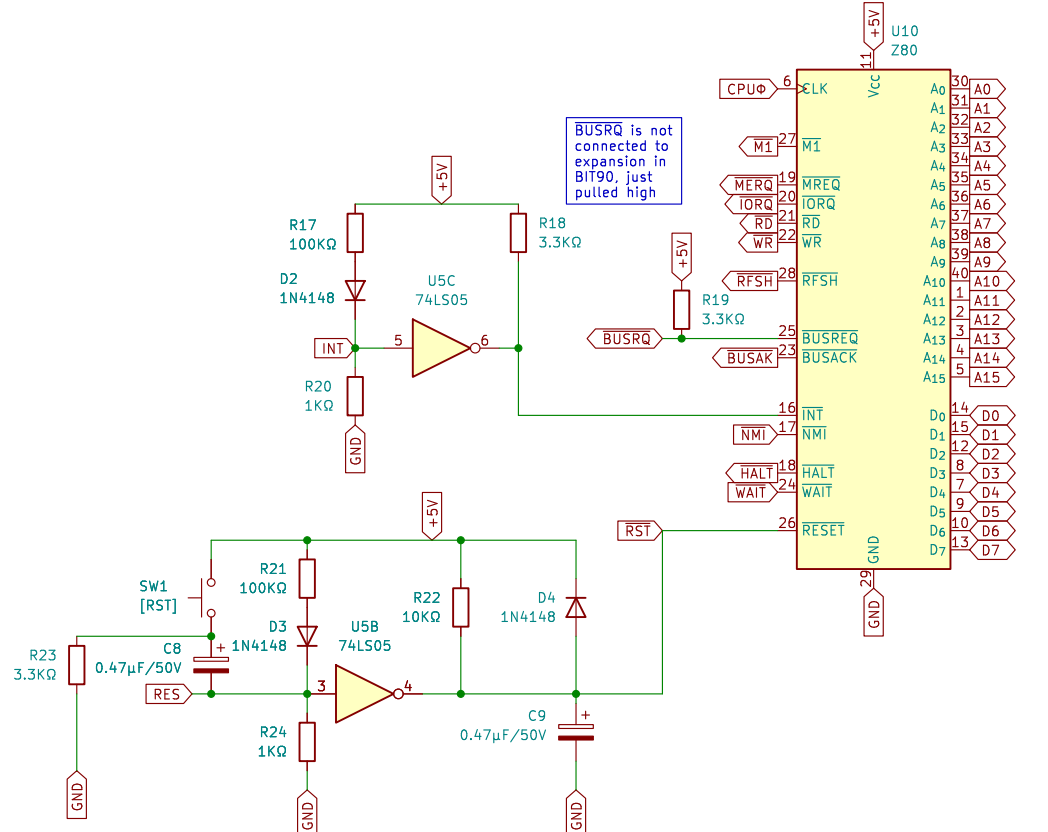


System Clocks



CPUΦ = CPU clock
VDPΦ = video clock
AUDΦ = audio clock (called EXTΦ on BIT90)
RΦ = current-limited CPU clock to expansion

CPU & Reset Circuit



Based on original schematics from BIT Corporation (普澤有限公司)
Modern recreation of the BIT90

Brett Hallen

Sheet: /Processor/
File: BIT90_26_Processor.kicad_sch

Title: BIT90/26 Computer – Processor

Size: A3 Date: 5/JUN/2026
KiCad E.D.A. 9.0.9

Rev: WIP
Id: 4/6

